

WHAT CAN LATENT WORLD MODELS KNOW? PHYSICAL INFORMATION IN MULTIMODAL PREDICTIVE REPRESENTATIONS

**Kaizhen Tan^{1,4,*} Sizhe Xu^{1,4,*} Xin Xu² Siru Tao² Yixiao Li²
Hanzhe Hong² Yang Feng³ Heqing Du³ Zhaonan Wang⁴**

¹New York University ²Carnegie Mellon University

³Columbia University ⁴NYU Shanghai

Project page: <https://tantansir.github.io/latent-world-model-identifiability/>

ABSTRACT

A central premise of latent world models is that predicting the future encourages representations to internalize the physics of their environment. We ask which physical quantities are accessible in learned latent states, how this depends on training, and how those quantities relate to the model’s predictions. We present POKEWORLD, a simulated environment in which a robot finger pushes objects whose mass, drag, and contact stiffness vary across episodes while remaining visually identical. We first measure parameter recovery from raw observation sequences, then train matched action-conditioned world models. Prediction targets strongly affect how well physical parameters can be decoded. For example, contact stiffness becomes decodable when touch is predicted, while providing touch only as an input does not. Across the tested settings, longer-horizon prediction improves linear estimation of position and velocity from learned representations relative to single-step prediction. Drag is recoverable from raw observations, but has weak linear readout from the learned latent states. Yet the models’ glide forecasts depend systematically on drag. Experiments on RH20T reproduce the same input–target dependence on real multimodal robot data. Together, these results show that observations, prediction targets, and prediction horizons shape both which physical quantities are accessible in latent states and how they affect future predictions.

1 INTRODUCTION

Latent world models learn a state representation by predicting how observations change under actions (LeCun, 2022; Hafner et al., 2025; Assran et al., 2025). Joint-embedding predictive architectures (JEPAs) make this prediction in a learned embedding space instead of pixel space (Assran et al., 2023; Balestriero & LeCun, 2025; Maes et al., 2026). For a robot, the value of such a state lies in the physics it captures: how heavy an object is, how quickly it slows down, and how hard it pushes back on contact. The prediction loss never names these quantities. Whether a trained latent contains them, and whether its forecasts depend on them, has to be measured.

Three difficulties make this measurement hard. First, a weak readout can reflect limited sensor information. Existing physical evaluations include video benchmarks (Garrido et al., 2025; Motamed et al., 2026) and probe-based readouts from multimodal robot encoders (Ye et al., 2026; Singh et al., 2026). These measurements alone do not establish what the observations make recoverable. Second, the training choices that could matter are rarely varied in isolation; models differ at once in which sensors they receive, which they predict, and how far ahead they predict. Third, readout and use are different properties. A probe can extract a parameter that the model’s forecasts ignore, and a forecast can depend on a parameter that no linear probe extracts.

*Equal contribution.

We address these difficulties with POKEWORLD, a simulated pushing environment in which objects look identical but differ from episode to episode in mass, drag, and contact stiffness. Because the simulator controls these parameters, we first measure how well each of them can be recovered from the raw observation streams, providing empirical recovery references. We then train action-conditioned JEPA world models that differ in one design choice at a time: the sensor streams they receive as inputs, the streams they predict as targets, and the horizons at which they predict (Fig. 1). Every frozen model is tested twice: whether a linear probe reads a parameter from its latent state, and whether its forecasts depend on that parameter. Finally, we test whether the input–target dependence carries over to real multimodal robot data.

Figure 1: **The world models, one column per time step.** Step s records a frame o_s , proprioception p_s , and touch τ_s (curves: finger speed, peak force), and a_s is the action applied during it; $\cdots$ marks omitted steps. $\theta = (m, \gamma, k)$ shapes the observations, not the pixels. The shared encoder E embeds every step, including the targets; the F variants also feed it p_s, τ_s . The causal transformer P reads z_s at position s , conditioned on a_s , and attends to earlier positions. Heads g predict embeddings $\Delta \in D = \{1, 4, 16\}$ steps ahead (for $\Delta > 1$ also from $a_{t+1:t+\Delta}$), each regressed onto the target below it; the X variants also predict p, τ at $t+1, t+4$. SIGReg acts on all z ; a linear probe W maps the frozen h_t to $\hat{\theta}$. In training every position predicts its own future.

Our contributions are as follows.

2 RELATED WORK

2025), and LeWorldModel trains such a model end to end from pixels (Maes et al., 2026); DreamerV3 combines reconstruction with latent dynamics (Hafner et al., 2025). Terver et al. (2026) study which design choices make JEPA world models plan well. Concurrent work restores state that a LeWorldModel latent omits with auxiliary targets (Zhu et al., 2026; Yan et al., 2026; Zeng et al., 2026; Liu et al., 2026c). Liu et al. (2026b) apply SIGReg to temporally centered residuals to preserve time-varying state information. JEPA-x shows that a state target can raise decodability without improving forecasts (Wen et al., 2026); Semigroup-JEPA conditions latent dynamics on given physical parameters (Liu et al., 2026a). We study hidden per-episode parameters and whether forecasts use them.

Probing physical knowledge. Intuitive-physics benchmarks evaluate video models (Garrido et al., 2025; Motamed et al., 2026), and CALIPER shows that frozen encoders decode mass- and friction-dependent outcomes from pixel displacement (Mehta & Baviskar, 2026); Joseph et al. (2026) locate and intervene on physical quantities inside video world models. Probe results depend on probe capacity (Hewitt & Liang, 2019; Belinkov, 2022). System identification asks whether a parameter is recoverable under a given observation and excitation process (Ljung, 1999), and recent theory states identifiability conditions for JEPA representations, action-conditioned transitions, and physical dynamics from video (Klindt et al., 2026; Zhang et al., 2026; Wang et al., 2026; Dong, 2026); other work analyzes JEPA risk and SIGReg and the link between prediction error and control (Cui et al., 2026; Arnez & Gomez-Villa, 2026; You et al., 2026). We pair each readout with a recovery reference from the same observations and with the model’s own forecasts.

Multimodal and physical estimation. Sparsh learns tactile representations (Higuera et al., 2025), TacGen improves contact-property decoding with tactile features (Ye et al., 2026), TouchWorld couples tactile prediction with action generation (Zhou et al., 2026), Kepler-Encoder fuses vision, proprioception, and force–torque (Singh et al., 2026), and HPT pretrains across embodiments (Wang et al., 2024). Video-based physical estimation recovers object parameters from observed interaction (Wu et al., 2015; 2016; Xu et al., 2019). Our factorial separates receiving a modality as input from predicting it as a target.

3 POKEWORLD

POKEWORLD hides physical parameters from every single frame and reveals them only through how the scene evolves (Fig. 2).

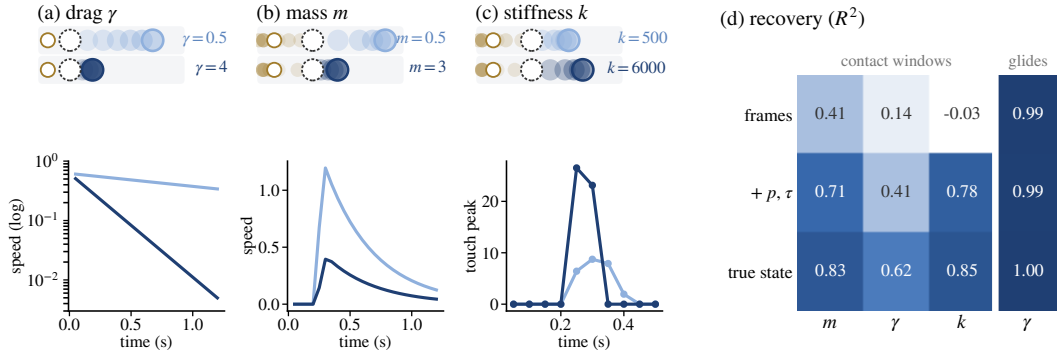


Figure 2: **Hidden physics in POKEWORLD.** (a)–(c) Matched episodes start from the same state and receive the same actions; only one parameter differs (light: low value, dark: high value). Top: the object every 0.2 s, so that spacing shows speed (dashed: start; gold: finger). Bottom: the stream in which the parameter shows. (d) Supervised recovery R^2 from 16-step windows of the frames (as the image-derived object trajectory), of the frames with proprioception and touch, and of the true object state, on contact windows and on pure glides. The supplementary material animates the three pairs.

Hidden physics. A force-controlled finger pushes one object in a two-dimensional arena. Each episode draws the object’s mass m , drag coefficient γ , and contact stiffness k , none of which affects

the rendering. The object’s velocity $\mathbf{v}$ follows

$$m \dot{\mathbf{v}} = \mathbf{f}_c - \gamma m \mathbf{v} + \mathbf{f}_w, \quad \mathbf{f}_c = k d^{3/2} \mathbf{n}, \quad (1)$$

where d is the finger–object overlap, $\mathbf{n}$ the contact normal, and $\mathbf{f}_w$ the wall force; light contact damping preserves impact transients (App. A). Each parameter leaves its trace in a different part of the motion (Fig. 2a–c). Drag sets the decay of a gliding object, $|\mathbf{v}(t)| = |\mathbf{v}(0)| e^{-\gamma t}$, independently of its mass. Mass sets the velocity change that a contact impulse produces. Stiffness sets the height and duration of the contact-force peak and changes the resulting motion only slightly.

Sensor streams. At every control step t the environment emits a 64×64 grayscale frame o_t , the finger’s proprioception p_t (position and velocity), touch τ_t (the mean, peak, and end-of-step reaction force within the step, the fraction of the step in contact, and a contact flag), and the finger force a_t commanded during the step. Sub-step touch statistics keep the impact peak that averaging over a step would erase. A mixed behavior policy produces sustained pushes, impacts, and free glides (App. A).

Empirical recovery reference. We measure recovery of a parameter θ_j from a set of streams by fitting a supervised estimator and evaluating it on held-out episodes. A two-layer GRU f_j reads 16-step windows x of the streams and actions, with the frames summarized by the image-derived object trajectory, and its held-out coefficient of determination

$$\rho_j^{\text{rec}} = 1 - \frac{\sum_i (\theta_{ij} - f_j(x_i))^2}{\sum_i (\theta_{ij} - \bar{\theta}_j)^2} \quad (2)$$

is an empirical recovery reference for θ_j : performance achieved by this GRU, not an information-theoretic upper bound. Recovery depends on the streams and context (Fig. 2d). Stiffness recovery improves with touch, and mass recovery improves with touch and proprioception. Drag recovery is lower on contact-dominated windows, even from the true object state, than on free glides, where frames alone give nearly exact recovery. Where recovery is high but latent readout is low, the tested probes access less of the parameter information available in the observations.

4 WORLD MODELS AND MEASUREMENTS

The accessibility of physical quantities in a world model may depend on what it is trained to predict. We therefore study action-conditioned JEPa world models that differ in three sets: the input streams I , the target streams U , and the prediction horizons D (Fig. 1).

Model. An encoder E embeds each step, and a causal transformer P with AdaLN action conditioning summarizes the history into the predictor state h_t :

$$z_t = E(o_t, \{u_t\}_{u \in I}), \quad h_t = P(z_{\leq t}, a_{\leq t}), \quad I \subseteq \{p, \tau\}. \quad (3)$$

Latent heads predict the embedding Δ steps ahead from h_t and the intervening actions, $\hat{z}_{t+\Delta} = g_\Delta(h_t, a_{t+1:t+\Delta})$, where the next-step head reads h_t alone; sensor heads predict the raw future values $\hat{u}_{t+\Delta}$ of every stream in a target set $U \subseteq \{p, \tau\}$. The model minimizes

$$\mathcal{L} = \sum_{\Delta \in D} w_\Delta \|\hat{z}_{t+\Delta} - z_{t+\Delta}\|^2 + \sum_{u \in U} \sum_{\Delta \in D_U} w_\Delta^u \|\hat{u}_{t+\Delta} - u_{t+\Delta}\|^2 + \lambda \text{SIGReg}(z), \quad (4)$$

where SIGReg keeps the embeddings close to an isotropic Gaussian and prevents collapse without a target network (Balestriero & LeCun, 2025; Maes et al., 2026). Unless stated otherwise, $D = \{1, 4, 16\}$ and $D_U = \{1, 4\}$ (App. B).

Variants. Model names encode I and U . V receives vision and predicts only embeddings ($I = U = \emptyset$). F adds proprioception and touch as inputs ($I = \{p, \tau\}$), and X adds them as targets ($U = \{p, \tau\}$; X_p and X_τ add one of them). VF thus receives every stream without predicting any, and VFX receives and predicts all of them. All variants share data, architecture, and compute.

Readout. For a frozen model, a ridge probe maps h_t at the end of a 16-step context to θ_j . Its held-out coefficient of determination, computed as in Eq. 2 on the same windows, is the readout ρ_j^{read} . The gap $\rho_j^{\text{rec}} - \rho_j^{\text{read}}$ compares recovery from the observations with linear readout from the latent; it does not by itself establish that the latent has discarded the parameter information.

Forecast dependence. A model can use a parameter that no linear probe extracts, or ignore one that a probe extracts. For a forecast $\hat{y}$ of a physical quantity, such as the object’s displacement or the peak contact force, we regress $\hat{y}$ and the simulator’s outcome y on θ_j , with the context state and the other parameters as covariates, and report the slope ratio

$$S = \beta_{\hat{y}|\theta_j} / \beta_{y|\theta_j}, \quad (5)$$

which is 1 when the forecast depends on θ_j as the physics does and 0 when it ignores θ_j . The continuation dependence S^{nat} uses the recorded future actions. The response dependence S^Δ applies two prescribed action sequences A_1 and A_0 from the same state and measures the dependence of the action effect $\Delta y = y(A_1) - y(A_0)$, with true outcomes from replaying the simulator. Its error

$$E^\Delta = \sum_i (\Delta \hat{y}_i - \Delta y_i)^2 / \sum_i \Delta y_i^2 \quad (6)$$

is 0 for exact effects and 1 for a model that predicts no effect.

Intervention models. To compare linear readout with forecast dependence, we train three models that receive the object’s position and velocity next to the images and predict with an interaction head, an MLP over h_t and the action window. The baseline uses VFX inputs and targets; the coordinate-target model adds auxiliary targets in which each parameter enters linearly (App. B), and the reference model receives the true parameters. Each test starts where the parameter matters: a push against a resting object for mass, a push timed to hit in the fourth frame for stiffness, and the release of a pushed object for drag.

Frozen-head diagnostic. To separate representation content from the objective used to train the forecast head, we freeze each predictor and train new heads H on the push test. A head receives h_t , the current object and finger state s_t , an action window A_b , and a parameter vector $c_\alpha = \hat{\theta} + \alpha(\theta - \hat{\theta})$, where $\hat{\theta}$ is a ridge estimate from h_t ; $\alpha = 0$ adds no parameter information beyond h_t and $\alpha = 1$ supplies the true parameters. With $\hat{d}_b$ the displacement that a fixed position probe decodes from $H(h_t, s_t, A_b, c_\alpha)$, the same head is trained on the same samples with either

$$\mathcal{L}_{\text{lat}} = \sum_{b \in \{0,1\}} \|H(h_t, s_t, A_b, c_\alpha) - z_{t+16}^{(b)}\|^2 \quad \text{or} \quad \mathcal{L}_{\text{eff}} = \|(\hat{d}_1 - \hat{d}_0) - (d_1 - d_0)\|^2, \quad (7)$$

where $z_{t+16}^{(b)}$ and d_b are the embedding and displacement that the simulator produces under A_b .

5 EXPERIMENTS

Setup. Models train on 4,000 POKEWORLD episodes of 64 steps. Probes are fitted and evaluated on separate episodes, and tests with prescribed actions use freshly simulated contexts stratified over the tested parameter (Apps. B and C). We report means over three training seeds; the appendix lists every seed.

Prediction targets shape parameter readout. Table 1 varies inputs and targets. Touch as a target, not as an input, makes stiffness decodable: VF receives the full touch stream and reads stiffness at chance, whereas VX_t, which predicts touch without receiving it, reads it at 0.43, and VFX at 0.50. Predicting proprioception instead leaves stiffness at chance, and removing the contact frames from the touch loss removes the stiffness readout of VFX (App. D). Mass follows the same order more weakly, and every model stays far below its recovery reference. Drag readout stays at about a third of its reference or less in these models; direct parameter supervision raises it above that reference (App. D).

Table 1: **Prediction targets affect physical-parameter readout.** Readout ρ^{read} (mean $\pm$ s.d. over three seeds) from the frozen predictor state on contact windows. The last two rows are the recovery references of Fig. 2d on the same windows.

Model	Inputs	Targets	$\log m$	γ	$\log k$
V	vision	latent	0.13 ± 0.01	0.07 ± 0.01	-0.03 ± 0.01
VF	vision, P, T	latent	0.21 ± 0.00	0.14 ± 0.01	-0.01 ± 0.00
VX_p	vision	latent, P	0.17 ± 0.01	0.09 ± 0.01	-0.01 ± 0.00
VX_t	vision	latent, T	0.25 ± 0.01	0.12 ± 0.01	0.43 ± 0.01
VX	vision	latent, P, T	0.26 ± 0.00	0.13 ± 0.01	0.47 ± 0.01
VFX	vision, P, T	latent, P, T	0.29 ± 0.01	0.12 ± 0.01	0.50 ± 0.01
Recovery from the VFX inputs			0.71	0.41	0.78
Recovery from the true object state			0.83	0.63	0.85

Table 2: **Multi-horizon prediction improves position and velocity readout.** Within each observation and regularization setting, vision-only V models differ only in their latent prediction heads (three seeds). Object position is read from the frame embedding and velocity from the predictor state; a_{t+1} is the action that produces the next frame. The three-head and multi-horizon rows share a summed loss weight of 2.

Latent prediction heads	grayscale, $\lambda = 0.1$		grayscale, $\lambda = 0.02$		frame + difference, $\lambda = 0.02$	
	position	velocity	position	velocity	position	velocity
one next-step head	0.05 ± 0.01	0.11 ± 0.02	0.85 ± 0.03	0.30 ± 0.01	0.67 ± 0.06	0.40 ± 0.01
one next-step head, given a_{t+1}	0.05 ± 0.01	0.14 ± 0.02	0.82 ± 0.03	0.33 ± 0.01	0.68 ± 0.04	0.40 ± 0.01
three next-step heads, given a_{t+1}	0.34 ± 0.23	0.18 ± 0.08	0.91 ± 0.02	0.39 ± 0.00	0.85 ± 0.05	0.43 ± 0.01
heads at $\Delta \in \{1, 4, 16\}$	0.90 ± 0.04	0.35 ± 0.04	0.99 ± 0.00	0.64 ± 0.01	0.95 ± 0.01	0.78 ± 0.01

Longer horizons improve motion readout. Table 2 compares vision-only models within each observation and regularization setting. A single next-step head yields weaker velocity readout and, under the stronger regularizer, weak position readout ($R^2 = 0.05$), while adding heads at $\Delta = 4$ and 16 improves both (position 0.90). Giving the next-step head the next action has little effect, and three next-step heads with the same total loss weight also yield lower velocity readout than the multi-horizon model. Longer horizons improve motion readout across the tested settings.

Drag is read on glides, and forecasts follow it. Drag shows only in the deceleration of a gliding object, which contact windows rarely contain. On pure glides the supervised GRU recovers drag almost exactly, and the VFX latent reads it far better than on contact windows, though still well below its reference (Fig. 3a). Its 16-step forecasts shorten with drag at the rate of the true dynamics (slope ratio 1.04), with a bias toward short displacements (Fig. 3b). Readability does not guarantee correct use: the coordinate-target model of Table 3 reads drag on the same glides almost perfectly, yet its forecast displacement grows with drag.

Readable is not the same as used. Table 3 compares linear readout and forecast dependence for the intervention models defined in Sec. 4. Coordinate targets make every parameter more readable at the start of its test, but the forecasts respond far less. The mass dependence of the predicted push effect reaches a quarter of the simulator’s, less than half of what the true-parameter model reaches, and its error does not fall; the drag dependence of the release forecast stays near zero. Stiffness is the exception, because peak force is itself a prediction target.

The objective of the head limits the response. On the coordinate-target predictor, $\mathcal{L}_{\text{eff}}$ nearly halves the error of $\mathcal{L}_{\text{lat}}$ at $\alpha = 0$ ($E^\Delta = 0.19$ versus 0.34) and approaches a supervised model given the true parameters, whereas $\mathcal{L}_{\text{lat}}$ barely improves even when it receives the true parameters (Fig. 4). The objective of the prediction head, more than the content of the representation, limits how accurately the model predicts what a new action will do.

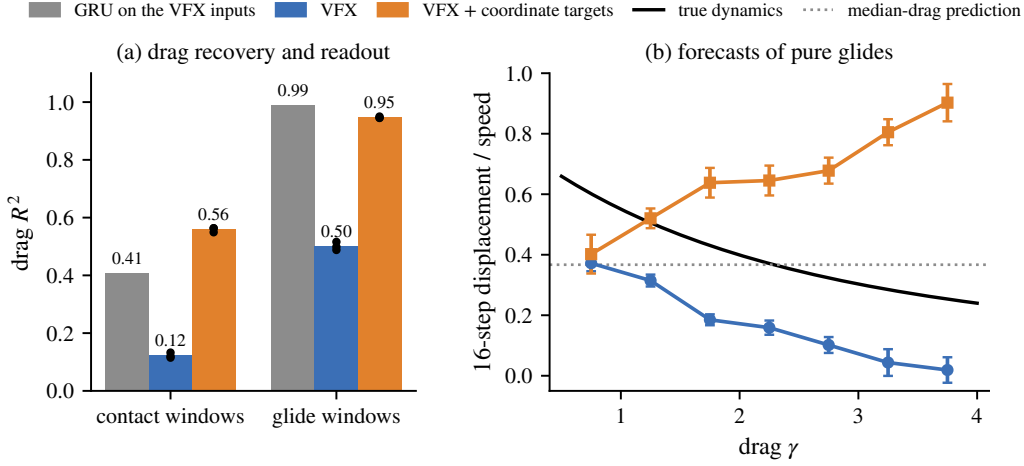


Figure 3: **Drag is recoverable, read on glides, and used by glide forecasts.** (a) Drag R^2 on contact windows and on pure glides for a GRU on the VFX inputs (recovery reference) and for ridge readout from the frozen predictor state of VFX and of the coordinate-target model defined in Sec. 4, which also receives the object state (points: seeds). (b) Predicted 16-step displacement divided by the speed at the end of the context, per drag bin (mean $\pm$ s.d. over three seeds), with the true dynamics and a prediction that assumes the median drag.

Table 3: **Readout versus forecast dependence.** Three models with object-state input and interaction heads: the VFX baseline, the same model with coordinate targets, and a model given the true parameters. Readout is measured at the start of each parameter’s test; S is the slope ratio of a forecast (Eq. 5; 1: as the physics) and E^Δ the error of the predicted action effect (Eq. 6). Means over three seeds (App. E).

Parameter	Measurement	Baseline	Coordinate targets	True parameters
mass m	readout	0.10	0.48	0.84
	S^{nat} , recorded continuation	0.19	0.36	0.98
	S^Δ , push vs. zero force	0.05	0.23	0.56
	E^Δ , push vs. zero force	0.68	0.72	0.65
stiffness k	readout	0.67	0.83	0.93
	S^{nat} , recorded continuation	0.68	0.74	0.97
	S^Δ , push vs. retreat	0.62	0.72	0.89
	E^Δ , push vs. retreat	0.33	0.31	0.28
drag γ	readout	0.06	0.41	0.70
	S , release	0.07	0.12	0.55

6 REAL MULTIMODAL ROBOT DATA

Setup. RH20T records robot manipulation with synchronized cameras, proprioception, and wrist force–torque (Fang et al., 2024). We train V, VX, VF, and VFX on the episodes of one robot and evaluate on tasks held out from training (App. F). The F variants add the end-effector pose and force–torque to a 96×96 grayscale view, and the X variants predict both. Because the recorded action is the next end-effector pose, we evaluate force, which the actions do not reveal: current force read from the frame embedding z_t , and force 0.4 s ahead read from the predictor state h_t .

Results. Table 4 reproduces the input–target dependence. Current-force readout is high for both the untrained fused-input encoder and VFX. Training VF with the same inputs on visual targets alone reduces readout R^2 below zero. Future force separates the models in the same way: VF falls

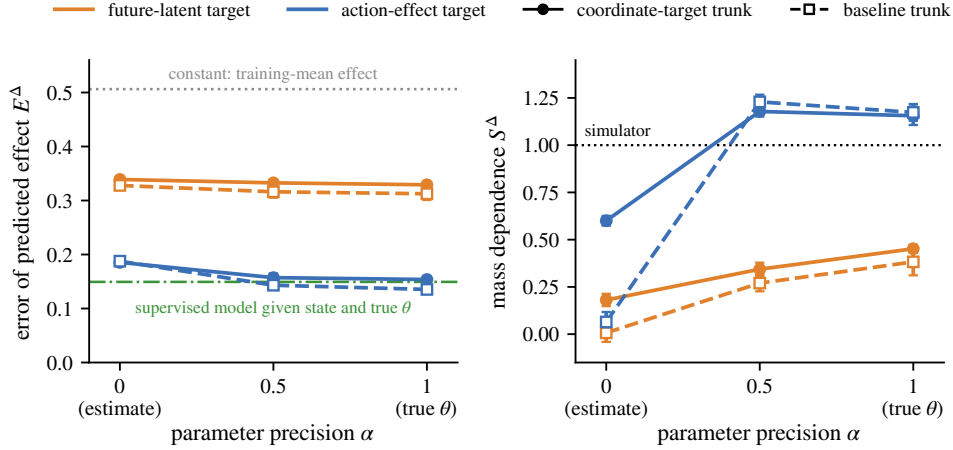


Figure 4: **The objective of the head limits the predicted action effect.** New heads on frozen predictors predict the effect of a 16-step push from the predictor state, the current object and finger state, the actions, and $c_\alpha = \hat{\theta} + \alpha(\theta - \hat{\theta})$, trained on future latents ($\mathcal{L}_{\text{lat}}$) or on the action effect ($\mathcal{L}_{\text{eff}}$). Left: error of the predicted effect. Right: its mass dependence. Error bars: s.d. over three trunk seeds.

Table 4: **Real robot data (RH20T, Flexiv): predicting force preserves its readout when force is included in the inputs.** Held-out-task R^2 (mean $\pm$ s.d. over three seeds) of current force read from the frame embedding and of force 0.4 s ahead read from the predictor state. The untrained model is the VFX network at initialization. For future force, a GRU trained on the raw sensor history reaches 0.46 and persistence 0.31.

Model	Inputs	Targets	Current force	Future force
V	image	latent	-0.28 ± 0.13	-0.15 ± 0.07
VX	image	latent, pose, force	-0.21 ± 0.02	-0.09 ± 0.06
VF	image, pose, force	latent	-0.16 ± 0.08	-0.10 ± 0.05
VFX	image, pose, force	latent, pose, force	0.88 ± 0.03	0.47 ± 0.00
untrained	image, pose, force	none	0.91	0.38

below the untrained model, while VFX matches a GRU trained directly on the raw sensor history. On a second robot, too, only VFX reads force on held-out tasks (App. F). Figure 5 follows force on two held-out episodes: the readout from the frozen VFX latent tracks a sustained press and repeated impulses, and its contact score rises before the measured onset.

7 DISCUSSION

Why prediction targets affect readout. Under the squared error of Eq. 4, the optimal head outputs the conditional mean of each target given h_t , so a parameter lowers the loss only if it moves this mean. Stiffness barely changes the object’s motion (Fig. 2c), so future embeddings hardly depend on it, whereas the touch peak does; stiffness readout improves when touch is a target and falls when contact frames are removed from the touch loss. Likewise, velocity barely changes the next frame but changes the frame sixteen steps later, and over a contact window the future depends far more on the push than on drag. Providing a stream as input alone does not ensure that its information remains accessible to the tested probes.

Implications for training world models. Three practices follow. A quantity whose readout matters can be included in the prediction targets. Predicting at several horizons improves motion readout in our experiments. A world model meant for control should be tested on the effects of actions it

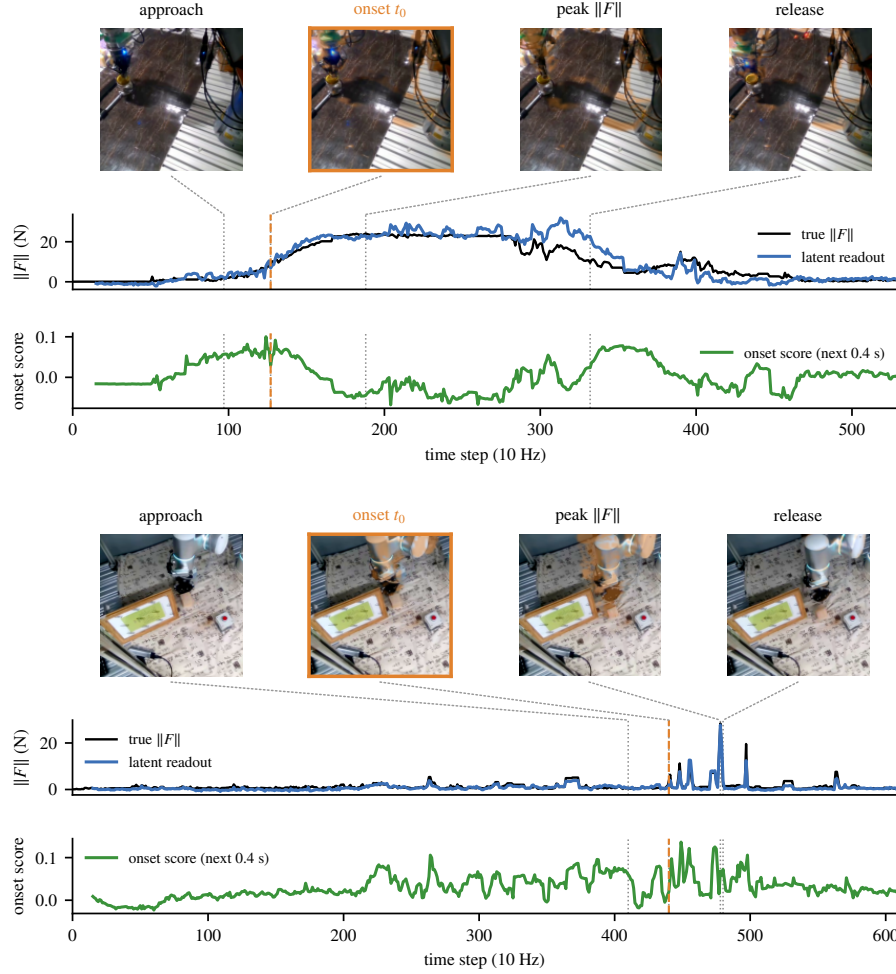


Figure 5: **Force read from the frozen VFX latent on held-out episodes** of two robots (top: KUKA; bottom: Flexiv). Each block shows four camera frames, measured force magnitude (black), force read from the latent (blue), and the readout score for contact within the next 0.4 s (green). The orange dashed line marks the measured contact onset. The model receives 96×96 grayscale images rather than the color crops shown.

has not seen, because neither readout nor continuation accuracy predicted how well our models answered a new action.

Limitations. POKEWORLD is two-dimensional with three hidden parameters, and our models have about five million parameters; larger visual world models may yield higher readout by default. The models in Table 3 and Fig. 4 receive the object state, and their coordinate targets are computed from simulator state. RH20T provides no parameter labels, so there we test force rather than hidden parameters. World models with explicit episode-level latent variables, which infer the parameters as part of prediction, are the natural next test.

8 CONCLUSION

Predicting the future does not by itself ensure high physical-parameter readout. In POKEWORLD, prediction targets shape parameter readout, and longer horizons improve position and velocity readout. Drag readout improves on glides, where forecasts follow drag at the rate of the true dynamics. Readability and use remain different properties: a model can read drag almost perfectly and still

forecast glides that lengthen with drag, and training the head on the action effect instead of on future latents nearly halves the error of a predicted push. The same input–target dependence holds on real robot data. These results support evaluating physical information through both linear readout and forecast dependence, under specified observations, targets, and prediction horizons.

AI USE STATEMENT

In this work, we used generative AI tools to help develop the conceptual framework of the study, propose and refine hypotheses, design and give feedback on experiments, implement methods (the simulator, the models, and the evaluations), clean and reformat the robot dataset, and interpret results. We have not used generative AI tools to generate data sets, to formulate mathematical claims, or for translation: every reported number comes from our simulator and from training and evaluating our models, and the paper states no theorems, so proofs and qualitative or thematic data analysis are not applicable. Additionally, we used generative AI tools to create and edit software code and scientific figures, suggest experimental parameters, search for, identify, and summarize literature, suggest the structure of the paper, draft and edit parts of the paper, and format references. We have reviewed all AI-assisted work: we tested the code, checked every reported number against the outputs of our experiments, checked interpretations against the results, and verified bibliographic information against the original publications. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

REPRODUCIBILITY STATEMENT

The environment, models, and evaluations run on a single GPU. The appendix lists every configuration, and code, data generators, and checkpoints will be released.

REFERENCES

- Fabio Arnez and Alexandra Gomez-Villa. The SIGReg objective as variational free energy: A theoretical active-inference account of JEPA world models. *arXiv preprint arXiv:2607.13612*, 2026. doi: 10.48550/arXiv.2607.13612. URL <https://arxiv.org/abs/2607.13612>.
- Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 15619–15629, 2023. doi: 10.1109/CVPR52729.2023.01499. URL https://openaccess.thecvf.com/content/CVPR2023/html/Assran_Self-Supervised_Learning_From_Images_With_a_Joint-Embedding_Predictive_Architecture_CVPR_2023_paper.html.
- Mido Assran, Adrien Bardes, David Fan, Quentin Garrido, Russell Howes, Mojtaba Komeili, Matthew Muckley, Ammar Rizvi, Claire Roberts, Koustuv Sinha, Artem Zhohus, Sergio Arnaud, Abha Gejji, Ada Martin, Francois Robert Hogan, Daniel Dugas, Piotr Bojanowski, Vasil Khalidov, Patrick Labatut, Francisco Massa, Marc Szafraniec, Kapil Krishnakumar, Yong Li, Xiaodong Ma, Sarath Chandar, Franziska Meier, Yann LeCun, Michael Rabbat, and Nicolas Ballas. V-JEPA 2: Self-supervised video models enable understanding, prediction and planning. *arXiv preprint arXiv:2506.09985*, 2025. doi: 10.48550/arXiv.2506.09985. URL <https://arxiv.org/abs/2506.09985>.
- Randall Balestriero and Yann LeCun. LeJEPA: Provable and scalable self-supervised learning without the heuristics. *arXiv preprint arXiv:2511.08544*, 2025. doi: 10.48550/arXiv.2511.08544. URL <https://arxiv.org/abs/2511.08544>.
- Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219, 2022. doi: 10.1162/coli_a.00422. URL https://doi.org/10.1162/coli_a.00422.
- Jingyi Cui, Qi Zhang, Hongwei Wen, and Yisen Wang. A generalization theory for JEPA-based world models. *arXiv preprint arXiv:2606.27014*, 2026. doi: 10.48550/arXiv.2606.27014. URL <https://arxiv.org/abs/2606.27014>.

- Yibin Dong. Why does the future branch? identifiable closure tests for stochastic physical world models. *arXiv preprint arXiv:2608.00591*, 2026.
- Hao-Shu Fang, Hongjie Fang, Zhenyu Tang, Jirong Liu, Chenxi Wang, Junbo Wang, Haoyi Zhu, and Cewu Lu. RH20T: A comprehensive robotic dataset for learning diverse skills in one-shot. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 653–660. IEEE, 2024. doi: 10.1109/ICRA57147.2024.10611615. URL <https://doi.org/10.1109/ICRA57147.2024.10611615>.
- Quentin Garrido, Nicolas Ballas, Mahmoud Assran, Adrien Bardes, Laurent Najman, Michael Rabbat, Emmanuel Dupoux, and Yann LeCun. Intuitive physics understanding emerges from self-supervised pretraining on natural videos. *arXiv preprint arXiv:2502.11831*, 2025. doi: 10.48550/arXiv.2502.11831. URL <https://arxiv.org/abs/2502.11831>.
- Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse control tasks through world models. *Nature*, 640(8059):647–653, 2025. doi: 10.1038/s41586-025-08744-2. URL <https://doi.org/10.1038/s41586-025-08744-2>.
- John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pp. 2733–2743. Association for Computational Linguistics, 2019. doi: 10.18653/v1/D19-1275. URL <https://aclanthology.org/D19-1275/>.
- Carolina Higuera, Akash Sharma, Chaithanya Krishna Bodduluri, Taosha Fan, Patrick Lancaster, Mrinal Kalakrishnan, Michael Kaess, Byron Boots, Mike Lambeta, Tingfan Wu, and Mustafa Mukadam. Sparsh: Self-supervised touch representations for vision-based tactile sensing. In Pulkit Agrawal, Oliver Kroemer, and Wolfram Burgard (eds.), *Proceedings of the 8th Conference on Robot Learning*, volume 270 of *Proceedings of Machine Learning Research*, pp. 885–915. PMLR, 2025. URL <https://proceedings.mlr.press/v270/higuera25a.html>.
- Sonia Joseph, Quentin Garrido, Randall Balestriero, Matthew Kowal, Thomas Fel, Shahab Bakhtiari, Blake Richards, and Michael Rabbat. Interpreting physics in video world models. In *Forty-third International Conference on Machine Learning (ICML)*, 2026.
- David Klindt, Yann LeCun, and Randall Balestriero. When does LeJEPa learn a world model? *arXiv preprint arXiv:2605.26379*, 2026.
- Yann LeCun. A path towards autonomous machine intelligence, 2022. URL <https://openreview.net/forum?id=BZ5alr-kVsF>. Version 0.9.2, June 27, 2022.
- Andy Zeyi Liu, Haoran Sun, Lucas Baker, Randall Balestriero, and John Sous. Semigroup-JEPa: Latent dynamics consistency for zero-shot physics generalization. *arXiv preprint arXiv:2609.10464*, 2026a.
- Chang Liu, Fei Suo, Yanzhou Jin, Zeyu Ping, Yusuke Iwasawa, Yutaka Matsuo, and Yaonan Zhu. Temporally centered SIGReg improves LeWorldModel representations for robot policy learning. *arXiv preprint arXiv:2607.26924*, 2026b.
- Muyuan Liu, Yue Huang, Zheng Liang, and Xiang Gao. Toward physically grounded JEPa world models for goal-conditioned robotic planning. *arXiv preprint arXiv:2609.03565*, 2026c.
- Lennart Ljung. *System Identification: Theory for the User*. Prentice Hall PTR, Upper Saddle River, NJ, 2nd edition, 1999. ISBN 978-0-13-656695-3.
- Lucas Maes, Quentin Le Lidec, Damien Scieur, Yann LeCun, and Randall Balestriero. LeWorld-Model: Stable end-to-end joint-embedding predictive architecture from pixels. *arXiv preprint arXiv:2603.19312*, 2026. doi: 10.48550/arXiv.2603.19312. URL <https://arxiv.org/abs/2603.19312>.
- Aman Mehta and Riya Baviskar. CALIPER: Clean scenes cannot rank physical inference in pre-trained visual representations. *arXiv preprint arXiv:2609.08250*, 2026.

- Saman Motamed, Laura Culp, Kevin Swersky, Priyank Jaini, and Robert Geirhos. Do generative video models understand physical principles? In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pp. 948–958, 2026. doi: 10.1109/WACV61042.2026.00099. URL https://openaccess.thecvf.com/content/WACV2026/html/Motamed_Do_Generative_Video_Models_Understand_Physical_Principles_WACV_2026_paper.html.
- Ishneet Sukhvinder Singh, Dhanoosh Pooranakumaran, Alex Nguyen, and Jia Qi Yip. Kepler-Encoder-v0.1: Towards a multimodal embedding model for robots. *arXiv preprint arXiv:2607.13522*, 2026. doi: 10.48550/arXiv.2607.13522. URL <https://arxiv.org/abs/2607.13522>.
- Basile Terver, Tsung-Yen Yang, Jean Ponce, Adrien Bardes, and Yann LeCun. What drives success in physical planning with joint-embedding predictive world models? *Transactions on Machine Learning Research*, 2026.
- Lirui Wang, Xinlei Chen, Jialiang Zhao, and Kaiming He. Scaling proprioceptive-visual learning with heterogeneous pre-trained transformers. In *Advances in Neural Information Processing Systems*, volume 37, pp. 124420–124450. Curran Associates, Inc., 2024. doi: 10.52202/079017-3952. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/e0f393e7980a24fd12fa6f15adfa25fb-Abstract-Conference.html.
- Yuanyuan Wang, Wenjie Wang, Kun Zhang, and Mingming Gong. Physics from video: Identifiability of time-invariant second-order ODEs under minimal trajectory conditions. In *International Conference on Machine Learning (ICML)*, 2026. arXiv:2606.00115.
- Kehan Wen, Ziming Li, Siyuan Luo, and Fan Shi. JEPA-x: Cross-predictive physics grounding for forecastable latent dynamics. *arXiv preprint arXiv:2608.24044*, 2026.
- Jiajun Wu, Ilker Yildirim, Joseph J. Lim, William T. Freeman, and Joshua B. Tenenbaum. Galileo: Perceiving physical object properties by integrating a physics engine with deep learning. In *Advances in Neural Information Processing Systems*, volume 28, pp. 127–135, 2015. URL <https://proceedings.neurips.cc/paper/2015/hash/d09bf41544a3365a46c9077ebb5e35c3-Abstract.html>.
- Jiajun Wu, Joseph J. Lim, Hongyi Zhang, Joshua B. Tenenbaum, and William T. Freeman. Physics 101: Learning physical object properties from unlabeled videos. In *Proceedings of the British Machine Vision Conference (BMVC)*, pp. 39.1–39.12, 2016. doi: 10.5244/C.30.39. URL <https://www.bmva-archive.org.uk/bmvc/2016/papers/paper039/index.html>.
- Zhenjia Xu, Jiajun Wu, Andy Zeng, Joshua B. Tenenbaum, and Shuran Song. DensePhysNet: Learning dense physical object representations via multi-step dynamic interactions. In *Robotics: Science and Systems XV*, 2019. doi: 10.15607/RSS.2019.XV.046. URL <https://www.roboticsproceedings.org/rss15/p46.html>.
- Haodong Yan, Jiaguan Zhu, Mingyuan Jia, Ruiqing Yin, Junjie He, Zhide Zhong, Junfeng Li, Jinxuan Lu, Hengtao Li, Tianran Zhang, Jiayi Chen, Wenxuan Song, Wen Chen, Yuxiang Gao, and Haoang Li. Is forward prediction enough? physical state grounding for JEPA world models. *arXiv preprint arXiv:2608.06799*, 2026.
- Wanghao Ye, Aarosh Das, Sihan Chen, Yiting Wang, Bowei Tian, Guoheng Sun, Shwai He, Zheyu Shen, Ziyao Wang, Yexiao He, Zhaoyi Liu, Meng Liu, Yuning Zhang, Meng Feng, Ziyi Wang, Yilong Dai, Yifei Dong, Siyuan Peng, Zhenle Duan, Joshua Liu, Lang Xiong, and Ang Li. TacGen: Touch is a necessary dimension of physical-world representation – addressing tactile data scarcity with scalable vision-to-touch alignment and generation. *arXiv preprint arXiv:2606.29173*, 2026. doi: 10.48550/arXiv.2606.29173. URL <https://arxiv.org/abs/2606.29173>.
- Hanzhe You, Yonggang Zhang, Maohao Ran, Zhiqin Yang, Zhenyuan Zhang, Wei Xue, Jun Song, Xinmei Tian, and Yike Guo. A control theory of predictability in latent world models. *arXiv preprint arXiv:2607.10362*, 2026. doi: 10.48550/arXiv.2607.10362. URL <https://arxiv.org/abs/2607.10362>.

Xi Zeng, Haojie Ren, and Ziying Song. PhyLatent: Learning dynamics-relevant representations for JEPA world models. *arXiv preprint arXiv:2608.05720*, 2026.

Xiangteng Zhang, Yang Guan, Bo Zhang, Hongyang Li, Ya-Qin Zhang, and Shengbo Eben Li. On the identifiability of controlled world models. *arXiv preprint arXiv:2607.22430*, 2026.

Gaoyue Zhou, Hengkai Pan, Yann LeCun, and Lerrel Pinto. DINO-WM: World models on pre-trained visual features enable zero-shot planning. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 79115–79135. PMLR, 2025. URL <https://proceedings.mlr.press/v267/zhou25t.html>.

Jianyi Zhou, Feiyang Hong, Yunhao Li, Yicheng Zhao, Yongjue Cen, Zirui Liu, Jiakang Huang, Zirui Chen, Ruiyang Zhang, Weizhuo Zhu, Xuhua Song, and Shuo Yang. TouchWorld: A predictive and reactive tactile foundation model for dexterous manipulation. *arXiv preprint arXiv:2607.07287*, 2026. doi: 10.48550/arXiv.2607.07287. URL <https://arxiv.org/abs/2607.07287>.

Penghao Zhu, Salvatore Penachio, Kaustav Mukherjee, and Aneesh Jonelagadda. Spectral-target physical latent structuring for JEPA-style world models. *arXiv preprint arXiv:2609.04264*, 2026.

A POKEWORLD DETAILS

Dynamics. Every episode samples $m \in [0.5, 3]$ and $k \in [500, 6000]$ log-uniformly and $\gamma \in [0.5, 4]$ uniformly. Bodies follow semi-implicit Euler integration at $dt=0.05$ with 20 substeps. The finger (mass 1, radius 0.06) is driven by a commanded force $F_{\max}a_t$ with $a_t \in [-1, 1]^2$. The object (radius 0.09) feels the drag force $-\gamma mv$, so its speed decays at rate γ regardless of mass. Finger–object contact follows the Hertzian penalty $F_c = k d^{3/2}$ with normal damping at 0.25 of critical, where d is the overlap; underdamping preserves the impact peak through which touch carries stiffness. The action a_t is the command applied during step t , so the state s_t is the result of a_t .

Sensors. Vision is an anti-aliased 64×64 grayscale frame (object intensity about 0.55, finger 1.0) together with its temporal difference. Proprioception is the finger’s position and velocity. Touch reports sub-step statistics of the reaction force on the finger in each control step: mean force (2-d), peak magnitude, end-of-step force (2-d), fraction of the step in contact, and a contact flag. Force channels carry noise with standard deviation 0.02. Sub-step statistics are necessary because averaging the force over a control step removes the impact peak.

Matched episodes. The pairs in Fig. 2a–c are replayed open loop from one initial state with one action sequence. The drag pair launches the object at speed 0.62 with the finger at rest ($m = 1.2$, $k = 2000$); the mass and stiffness pairs strike a resting object with five steps of full force followed by a retreat ($\gamma = 2.5$; $k = 2000$ for the mass pair, $m = 1.2$ for the stiffness pair).

Behavior policy and data. Episodes of 64 steps mix pursuit, strike-and-retreat, and Ornstein–Uhlenbeck drift with random switching, and a fraction of episodes launch the object with a random velocity. The training set has 4,000 episodes; probes are fitted on 1,600 and evaluated on 800 independent episodes. Tests with prescribed actions draw contexts from 48,000 freshly simulated episodes per test.

B MODEL AND TRAINING DETAILS

Architecture. The per-frame encoder is a four-stage CNN (channels 32/64/128/256, stride 2) over the image channels, with MLPs for proprioception and touch, fused by a linear projection with BatchNorm into a 128-dimensional embedding; following Maes et al. (2026), there is no trailing LayerNorm. The predictor is a four-layer causal transformer (width 192, eight heads, dropout 0.1) with zero-initialized AdaLN action conditioning. The next-step head is a linear projection of h_t with BatchNorm. Heads at $\Delta \in \{4, 16\}$ are additive in h_t and an MLP embedding of the flattened action window, $\hat{z}_{t+\Delta} = W_h h_t + W_a \phi(a_{t+1:t+\Delta}) + b$. Cross-modal heads predict raw touch and

proprioception at $t+1$ from h_t and at $t+4$ with $a_{t+1:t+4}$. Loss weights are 1 for the next-step latent, 0.5 for the other latent heads and the next-step cross-modal heads, and 0.25 for the four-step cross-modal heads; contact frames carry weight 4 in the touch loss.

Optimization. AdamW with learning rate 3×10^{-4} , cosine decay, 500 warmup steps, and weight decay 0.05; batches of 128 windows of 24 frames; 20,000 steps in bf16. SIGReg has weight $\lambda = 0.02$ unless stated otherwise and uses 1024 random directions and 17 integration knots, resampled at every step. A model trains in about seven minutes on one H200 GPU.

Models of Table 3. The object’s position and velocity enter the encoder next to the image features. Interaction heads replace the additive heads with a two-layer MLP (width 256) over h_t and the action embedding. The coordinate targets are predicted from the predictor state with loss weight 0.5 each: for drag, the four-step change $\log |v_{t+4}| - \log |v_t| = -4\gamma dt$ on contact-free windows in which the object keeps moving; for stiffness, $\log |f_{t+1}| - 1.5 \log d_{t+1} = \log k$ on contact frames; for mass, $\log |\bar{f}_{t+1}| - \log a_{\parallel} = \log m$ on contact frames away from the walls, where $a_{\parallel}$ is the object’s acceleration along the force plus the drag term $\gamma \bar{v}$. The true-parameter model receives $(\log m, \gamma, \log k)$ as additional conditioning of every predictor layer.

Control variants. The reconstruction control replaces the latent prediction losses with a deconvolutional pixel decoder (four stride-2 upconvolutions from a 4×4 map, channels 256/128/64/32) and removes SIGReg; cross-modal losses stay active. The parameter-supervision control adds a two-layer head from h_t to the whitened $(\log m, \gamma, \log k)$ with weight 0.5. The Gaussian-NLL control adds variance heads to every latent prediction.

C MEASUREMENT DETAILS

Windows and probes. Probe windows are 16-step contexts starting at steps 0, 8, $\dots$, 40 of probe episodes. Contact windows contain at least two contact frames (3,228 test windows); moving windows have a mean object speed above 0.15. Ridge probes standardize the features and select the penalty on a held-out part of the fit episodes; MLP probes are early-stopped two-layer MLPs. Object position is read from the frame embedding z_t and object velocity from h_t on moving windows.

Recovery references. The GRU references (two layers, width 96) read 16-step windows and stop early on 15% of the fit episodes. Image-derived trajectories use the object’s intensity band after removing the finger disk, which is located from the image, with central-difference velocities. Table 5 lists the references on the probe windows. Contact windows rarely contain a free glide, so for drag we also use the pure-glide windows of Fig. 3 (one window per episode, stratified over γ) with five-fold cross-validation over episodes. There, the GRU recovers drag at 0.99 from the frames alone, at 0.99 from the VFX inputs, and at 1.00 from the true object state (Fig. 2d), and the structured estimator $\hat{\gamma} = -\text{median}_t \log(|v_{t+1}|/|v_t|)/dt$ reaches 0.96 from image-derived speeds and 1.00 from simulator speeds.

Table 5: **Recovery references.** Test R^2 for $\log m / \gamma / \log k$ by input set and window class; GRU, with ridge regression on the flattened window in parentheses. V: image-derived object position and velocity; P: proprioception; T: touch; A: actions.

Windows	Inputs	$\log m$	γ	$\log k$
contact	V+A	0.41 (−0.01)	0.14 (0.00)	−0.03 (0.00)
contact	V+P+T+A	0.71 (0.03)	0.41 (−0.01)	0.78 (0.20)
contact	object state+P+T+A	0.83 (0.03)	0.63 (−0.01)	0.85 (0.20)
all	V+A	0.24 (−0.01)	0.15 (0.00)	−0.05 (0.00)
all	V+P+T+A	0.45 (0.02)	0.41 (−0.01)	0.49 (0.13)
all	object state+P+T+A	0.52 (0.02)	0.56 (−0.01)	0.51 (0.13)

Tests with prescribed actions. Mass contexts end with the object slower than 0.10 and out of contact for two frames, contain at least two contact frames, and have the finger 0.16–0.45 from the

object, away from the walls. The push applies 0.9 unit(object – finger) for 16 steps; the alternative applies zero commanded force, and the finger then moves under the simulator dynamics. Stiffness contexts end with the finger 0.02–0.10 from the object after at least two contact frames; push and retreat apply ± 0.9 along the same direction for four steps, and we keep starts where the push first makes contact in the fourth frame and the retreat never does. Drag contexts end with the object in contact for two frames and moving faster than 0.2 with room ahead; the finger then withdraws for 16 steps. Contexts are stratified over the tested parameter with at most 300 in each of six bins, giving 1,788 mass, 1,455 stiffness, and 1,800 drag contexts. Slopes control for the finger–object distance, the finger speed, the finger and object velocities along the action direction, the direction, and the other two parameters (for stiffness: gap, normal and tangential relative velocity, direction, $\log m$, γ). Mass and drag forecasts decode the $\Delta=16$ prediction with a linear position probe; stiffness forecasts use the peak-force channel of the $\Delta=4$ touch head. Readout at the start of a test is an episode-grouped ridge probe on the same contexts.

Frozen-head diagnostic. Mass contexts are drawn from independent episode sets for training (96,000 episodes, 5,426 starts), validation (1,500 starts), and testing (the 1,788 starts above). Each start stores h_t , the current object and finger state, and the replayed outcomes of both branches: 16-step displacements d_1, d_0 and the frozen encoder’s embeddings z_1, z_0 of the true frame 16. $\hat{\theta}$ is a ridge regression from h_t to the standardized parameters, cross-fitted over five episode folds on the training starts; its test R^2 for $(\log m, \gamma, \log k)$ is 0.51/0.44/0.62 on the coordinate-target predictor and 0.12/0.03/0.31 on the baseline. The head is a two-layer MLP (width 256) from $(h_t, s_t, \phi(A), c_\alpha)$ to an embedding decoded by the fixed position probe. The latent target is the squared error to z_1 and z_0 ; the action-effect target is the squared error of $\hat{d}_1 - \hat{d}_0$ to $d_1 - d_0$. Both use the same samples, initialization, optimizer (AdamW, 3×10^{-4} , 4,000 steps), and validation-based selection. The supervised reference is an MLP on $(s_t, \theta, \text{direction})$ trained on the same starts. Per-trunk values are reported in App. E.

D ADDITIONAL POKEWORLD RESULTS

Nonlinear probes. Early-stopped MLP probes on the same contact windows reproduce the ridge results of Table 1 (Table 6).

Table 6: **MLP probes.** Early-stopped two-layer MLP readout (R^2 , mean $\pm$ s.d. over three seeds) from the frozen predictor state on contact windows.

Model	$\log m$	γ	$\log k$
V	0.00 $\pm$ 0.00	−0.01 $\pm$ 0.00	−0.00 $\pm$ 0.00
VF	0.00 $\pm$ 0.00	−0.00 $\pm$ 0.00	−0.00 $\pm$ 0.00
VX _p	−0.00 $\pm$ 0.00	−0.01 $\pm$ 0.00	−0.00 $\pm$ 0.00
VX _t	0.20 $\pm$ 0.01	0.04 $\pm$ 0.01	0.42 $\pm$ 0.00
VX	0.22 $\pm$ 0.01	0.06 $\pm$ 0.01	0.45 $\pm$ 0.00
VFX	0.26 $\pm$ 0.02	0.05 $\pm$ 0.01	0.50 $\pm$ 0.01

Objective controls. Keeping the VFX predictor, data, and probes fixed, we change only the objective (Table 7). Supervising the parameters directly raises every parameter readout, most for stiffness and mass; pixel reconstruction in place of latent prediction leaves drag readout low, and stiffness remains decodable with touch forecasting.

Table 7: **Objective controls** (one seed). Ridge readout from the predictor state on contact windows; position from the frame embedding; velocity on moving windows.

Objective	$\log m$	γ	$\log k$	position	velocity
VFX	0.30	0.11	0.49	0.98	0.78
VFX + parameter supervision	0.73	0.46	0.92	0.97	0.76
VFX, Gaussian NLL	0.30	0.12	0.46	1.00	0.88
VFX, pixel reconstruction	0.18	0.04	0.57	0.95	0.65
V, pixel reconstruction	0.16	0.08	-0.01	0.69	0.50

Regularizer weight. A lighter SIGReg weight sharpens every readout of the VFX model (Table 8).

Table 8: **SIGReg weight** (VFX, one seed). Ridge readout on contact windows; velocity on moving windows.

λ	0.3	0.1	0.02	0.01	0.005
$\log k$	0.13	0.30	0.49	0.51	0.57
$\log m$	0.14	0.19	0.30	0.32	0.32
γ	0.00	0.05	0.11	0.13	0.12
velocity	0.54	0.63	0.78	0.78	0.80

Touch-loss weight on contact frames. Stiffness readout depends on how strongly the touch loss weighs contact frames, while mass and drag barely move (Table 9).

Table 9: **Contact-frame weight of the touch loss** (VFX, one seed; the default is 4). Ridge readout on contact windows.

weight	4	1	0.25	0
$\log k$	0.49	0.31	0.10	0.00
$\log m$	0.30	0.28	0.26	0.22
γ	0.11	0.13	0.14	0.15

E PER-SEED RESULTS FOR READOUT VERSUS FORECAST DEPENDENCE

The additive heads of Table 1 give an action effect $W_a[\phi(A_1) - \phi(A_0)]$ that does not depend on the history, and their mass dependence of the push effect is 0.01, 0.00, and -0.01 in the three seeds. The tables below list every seed, the direct-supervision and self-conditioning variants, and the frozen-head results of Fig. 4.

Table 10: **Mass and drag tests, per seed.** Every slash separates the three representation seeds. Columns: mass push-branch slope ratio, mass action-effect ratio S^Δ and relative error E^Δ , mass continuation ratio S^{nat} , and drag release-branch slope ratio. Models: additive = VFX with the additive head and no state input; interaction, no state = interaction head without state input; baseline, coordinate, oracle = the three models of Table 3 (oracle: true parameters); + self-cond. = coordinate targets with the model’s own coordinate estimates fed to the predictor; direct sup. = parameter-supervision heads, no state input.

model	m push	m S^Δ	m E^Δ	m S^{nat}	γ release
additive	0.06/0.03/-0.01	0.01/-0.00/-0.01	0.54/0.52/0.54	0.03/0.00/-0.04	-0.01/-0.03/-0.06
interaction, no state	0.03/0.08/0.09	0.07/0.09/0.04	0.85/0.80/0.90	0.14/0.14/0.12	-0.01/-0.00/-0.03
baseline	0.06/0.11/0.12	0.04/0.00/0.11	0.64/0.77/0.64	0.16/0.19/0.23	0.06/0.10/0.05
coordinate	0.37/0.29/0.41	0.22/0.13/0.35	0.73/0.75/0.69	0.37/0.36/0.35	0.10/0.15/0.12
+ self-cond.	0.34/0.29/0.37	0.21/0.19/0.22	0.66/0.63/0.76	0.35/0.33/0.40	0.08/0.11/0.11
direct sup.	0.34/0.43/0.29	0.27/0.39/0.28	1.01/1.01/0.89	0.26/0.33/0.32	-0.00/0.00/0.01
oracle	1.12/1.23/1.04	0.58/0.71/0.39	0.63/0.69/0.62	0.98/0.97/0.98	0.54/0.55/0.56

Table 11: **Stiffness test, per seed** (slashes separate seeds). Push-branch slope ratio, action-effect ratio S^Δ and relative error E^Δ (push minus retreat), continuation ratio S^{nat} , and the stiffness readout at the impact start.

model	k push	k S^Δ	k E^Δ	k S^{nat}	k readout at start
additive	0.39/0.39/0.40	0.00/0.00/0.00	1.00/1.00/1.00	0.53/0.53/0.54	0.37/0.41/0.44
interaction, no state	0.59/0.59/0.56	0.60/0.61/0.54	0.39/0.38/0.38	0.63/0.66/0.61	0.62/0.61/0.61
baseline	0.62/0.62/0.62	0.63/0.61/0.61	0.35/0.32/0.30	0.67/0.68/0.68	0.65/0.67/0.69
coordinate	0.69/0.63/0.70	0.75/0.69/0.73	0.31/0.32/0.30	0.77/0.69/0.77	0.81/0.83/0.85
+ self-cond.	0.76/0.66/0.72	0.78/0.68/0.74	0.25/0.25/0.24	0.79/0.66/0.74	0.86/0.83/0.85
direct sup.	0.69/0.70/0.67	0.70/0.72/0.65	0.39/0.36/0.37	0.74/0.75/0.73	0.86/0.87/0.87
oracle	0.87/0.88/0.88	0.89/0.89/0.88	0.30/0.29/0.25	0.96/0.97/0.97	0.94/0.93/0.93

Table 12: **Readouts of the same checkpoints** (three-seed means). Readout at the intervention start is the episode-grouped ridge R^2 on the mass-push contexts (m), the release contexts (γ), and the impact contexts (k); contact and glide columns are the ridge probes of Table 1 on the standard probe windows. The oracle is given the parameters.

model	start m	start γ	start k	contact m	contact γ	contact k	glide γ
additive	0.03	0.05	0.41	0.29	0.12	0.50	0.13
interaction, no state	0.07	0.05	0.61	0.38	0.13	0.64	0.08
baseline	0.10	0.06	0.67	0.45	0.14	0.65	0.08
coordinate	0.48	0.41	0.83	0.76	0.56	0.82	0.82
+ self-cond.	0.50	0.40	0.84	0.78	0.55	0.83	0.82
direct sup.	0.43	0.25	0.87	0.75	0.47	0.92	0.43
oracle	0.84	0.70	0.93	0.76	0.49	0.85	0.43

Heads trained on future latents with the actions alone (Table 13) reproduce the pattern of Fig. 4: true parameters raise the mass dependence of the push effect but not its accuracy, and an estimate extracted from the same representation adds no more than a random projection of it. Table 14 gives the results for the heads in Fig. 4, which also receive the current object and finger state.

Table 13: **New heads on frozen predictors, trained on future latents.** $H(h, A)$ uses the predictor state and the actions; $H(h, A, \hat{\theta})$ adds a cross-fitted ridge estimate of the parameters from h ; $H(h, A, Ph)$ adds a fixed random three-dimensional projection of h instead; $H(h, A, \theta)$ adds the true parameters. Mass push-branch slope ratio, mass dependence S^Δ and error E^Δ of the push effect, and drag release slope; means over three seeds.

Frozen predictor	Head	m push	m S^Δ	m E^Δ	γ release
baseline	$H(h, A)$	0.12	0.02	0.68	0.10
	$H(h, A, \hat{\theta})$	0.14	0.12	0.67	0.09
	$H(h, A, Ph)$	0.12	0.09	0.67	0.09
	$H(h, A, \theta)$	0.68	0.65	0.66	0.07
coordinate targets	$H(h, A)$	0.39	0.25	0.66	0.13
	$H(h, A, \hat{\theta})$	0.43	0.32	0.71	0.12
	$H(h, A, Ph)$	0.41	0.27	0.67	0.12
	$H(h, A, \theta)$	0.91	0.76	0.68	0.12

Table 14: **Training target versus parameter precision on the mass action effect.** New heads on frozen trunks receive the trunk state, the current object and finger state, the action window, and $c_\alpha = \hat{\theta} + \alpha(\theta - \hat{\theta})$, and are trained with either the latent target or the action-effect target on the same intervention samples; $\alpha = 0$ adds no test-time parameter information. Columns: action-effect slope ratio S^Δ , relative error E^Δ , RMSE (arena units), and skill relative to the training-mean effect, on 1,788 held-out test starts; mean $\pm$ s.d. over three trunk seeds with the head-training seed fixed. The constant and the true-parameter control do not depend on the trunk (fixed).

head	S^Δ	E^Δ	RMSE	skill
coordinate-target trunk (3 seeds)				
constant, training-mean effect (fixed)	-0.00	0.51	0.108	0.00
supervised control, $(s_t, \theta, \text{dir})$ (fixed)	1.07	0.15	0.058	0.71
supervised control, $(s_t, \hat{\theta}, \text{dir})$	0.56 $\pm$ 0.00	0.17 $\pm$ 0.00	0.062 $\pm$ 0.001	0.67 $\pm$ 0.01
latent target, $\alpha = 0$	0.18 $\pm$ 0.03	0.34 $\pm$ 0.00	0.088 $\pm$ 0.001	0.33 $\pm$ 0.01
latent target, $\alpha = 0.5$	0.34 $\pm$ 0.03	0.33 $\pm$ 0.00	0.087 $\pm$ 0.000	0.34 $\pm$ 0.01
latent target, $\alpha = 1$	0.45 $\pm$ 0.02	0.33 $\pm$ 0.00	0.087 $\pm$ 0.000	0.35 $\pm$ 0.01
action-effect target, $\alpha = 0$	0.60 $\pm$ 0.03	0.19 $\pm$ 0.01	0.065 $\pm$ 0.001	0.63 $\pm$ 0.01
action-effect target, $\alpha = 0.5$	1.18 $\pm$ 0.03	0.16 $\pm$ 0.01	0.060 $\pm$ 0.001	0.69 $\pm$ 0.02
action-effect target, $\alpha = 1$	1.16 $\pm$ 0.05	0.15 $\pm$ 0.01	0.059 $\pm$ 0.001	0.70 $\pm$ 0.01
baseline trunk (3 seeds)				
supervised control, $(s_t, \hat{\theta}, \text{dir})$	0.05 $\pm$ 0.06	0.19 $\pm$ 0.01	0.065 $\pm$ 0.001	0.63 $\pm$ 0.01
latent target, $\alpha = 0$	0.01 $\pm$ 0.05	0.33 $\pm$ 0.01	0.087 $\pm$ 0.001	0.35 $\pm$ 0.01
latent target, $\alpha = 0.5$	0.27 $\pm$ 0.04	0.32 $\pm$ 0.01	0.085 $\pm$ 0.001	0.38 $\pm$ 0.02
latent target, $\alpha = 1$	0.38 $\pm$ 0.07	0.31 $\pm$ 0.01	0.085 $\pm$ 0.002	0.38 $\pm$ 0.02
action-effect target, $\alpha = 0$	0.06 $\pm$ 0.05	0.19 $\pm$ 0.01	0.066 $\pm$ 0.001	0.63 $\pm$ 0.01
action-effect target, $\alpha = 0.5$	1.23 $\pm$ 0.04	0.14 $\pm$ 0.00	0.057 $\pm$ 0.000	0.72 $\pm$ 0.00
action-effect target, $\alpha = 1$	1.17 $\pm$ 0.04	0.14 $\pm$ 0.00	0.056 $\pm$ 0.000	0.73 $\pm$ 0.00

F RH20T DETAILS

Data and preprocessing. We use the Flexiv configuration of RH20T (4,258 episodes; 2,947 training episodes from 112 tasks; 479 evaluation episodes from 12 held-out tasks) and the KUKA configuration (896 episodes, 56 tasks) at 10 Hz. Models receive a 96×96 grayscale view from one camera; the F variants add the end-effector pose and the six-axis force-torque signal. Streams are clipped to the 0.1–99.9 percentiles of the training episodes and whitened. The 800-episode subset is sampled per task and has its own clipping and whitening statistics; the future-force label, the persistence prediction, and the evaluation windows are identical at both sizes. The recorded action

at step t is the pose at $t+1$; for future-force readout the predictor state is computed with that last action replaced by the previous one, and the raw-history GRU receives the same history. Contact onset marks $\|F\| > 10$ N within the next four steps while $\|F_t\| < 10$ N. The future-force readout is a two-layer MLP (256–128 units) with early stopping, fitted on training-task probe episodes.

Table 15: **Flexiv, all metrics** (mean $\pm$ s.d. over three seeds; IID: held-out episodes of training tasks; OOD: held-out tasks). Force and position are read from the frame embedding; contact-onset AUC from the predictor state. The untrained VFX network is evaluated at initialization.

Model	Episodes	force IID	force OOD	position IID	position OOD	onset AUC OOD
V	800	-0.05 ± 0.01	-0.23 ± 0.01	0.19 ± 0.04	0.11 ± 0.07	0.68 ± 0.01
	2,947	-0.06 ± 0.01	-0.28 ± 0.13	0.16 ± 0.09	0.03 ± 0.12	0.71 ± 0.03
VX	800	-0.03 ± 0.01	-0.22 ± 0.07	0.31 ± 0.03	0.29 ± 0.03	0.71 ± 0.04
	2,947	-0.04 ± 0.01	-0.21 ± 0.02	0.30 ± 0.00	0.23 ± 0.02	0.76 ± 0.05
VF	800	0.15 ± 0.02	-0.08 ± 0.05	0.96 ± 0.01	0.96 ± 0.01	0.83 ± 0.03
	2,947	0.14 ± 0.04	-0.16 ± 0.08	0.96 ± 0.01	0.96 ± 0.01	0.85 ± 0.01
VFX	800	0.93 ± 0.00	0.90 ± 0.01	0.97 ± 0.00	0.97 ± 0.00	0.90 ± 0.01
	2,947	0.92 ± 0.01	0.88 ± 0.03	0.97 ± 0.00	0.97 ± 0.00	0.88 ± 0.00
untrained VFX	none	0.94	0.91	0.96	0.98	0.88

Data scale. With $3.7\times$ more training episodes, future-force readout of VFX rises from 0.40 to 0.47 on held-out tasks in every seed, while VF stays below zero at both sizes (Table 16). The GRU reference reads the same 16-step history as the models, and an MLP on the current force and pose alone reaches 0.38.

Table 16: **Flexiv future force** (0.4 s ahead; R^2 , mean $\pm$ s.d. over three seeds), with IID and held-out-task (OOD) splits.

	800 episodes		2,947 episodes	
	IID	OOD	IID	OOD
V	-0.01 ± 0.01	-0.10 ± 0.04	-0.03 ± 0.02	-0.15 ± 0.07
VX	0.07 ± 0.03	-0.07 ± 0.04	0.08 ± 0.05	-0.09 ± 0.06
VF	0.09 ± 0.03	-0.19 ± 0.06	0.09 ± 0.02	-0.10 ± 0.05
VFX	0.39 ± 0.01	0.40 ± 0.02	0.43 ± 0.01	0.47 ± 0.00
untrained model, fused inputs	0.37	0.38	0.37	0.38
GRU on raw 16-step history	0.44	0.47	0.44	0.46
MLP on current force and pose	0.40	0.38	0.40	0.38
persistence	0.23	0.31	0.23	0.31

KUKA. On the KUKA configuration (single seed; clipping thresholds computed over all episodes), only VFX reads force on held-out tasks (Table 17).

Table 17: **KUKA** (one seed). Force and position readout from the frame embedding and contact-onset AUC (IID / held-out tasks).

Model	force	position	onset AUC
V	$-0.05 / -0.49$	$0.76 / 0.48$	$0.70 / 0.54$
VX_t	$-0.04 / -0.61$	$0.77 / 0.46$	$0.80 / 0.72$
VX	$-0.02 / -0.70$	$0.75 / 0.50$	$0.77 / 0.69$
VFX	$0.87 / 0.56$	$0.98 / 0.98$	$0.86 / 0.70$